

Task #1: Restore DB from the Snapshot

1. Created the RDS database:

The screenshot shows the AWS RDS console for a database instance named 'database-1'. The 'Summary' tab is active, displaying the following details:

DB identifier	Status	Role	Engine	Recommendations
database-1	Available	Instance	MySQL Community	

CPU	Class	Current activity	Region & AZ
3.92%	db.t4g.micro	0 Connections	us-east-1a

Below the summary, the 'Connectivity & security' tab is selected. It shows the following configuration:

- Endpoint & port:** Endpoint is `database-1.c2dg6mmy4rhh.us-east-1.rds.amazonaws.com`, Port is `3306`.
- Networking:** Availability Zone is `us-east-1a`, VPC is `vpc-06008718ac30863d8`, Subnet group is `default-vpc-06008718ac30863d8`. Subnets include `subnet-074100db9c88d1d1b`, `subnet-0d52e0a0fa761063c`, `subnet-0788c08d2a5c09f64`, `subnet-0f2c46afa395150ec`, and `subnet-055dd26c1ec73d43f`.
- Security:** VPC security groups include `default (sg-070dc7952b1b5af5f)` (Active). Publicly accessible is `Yes`. Certificate authority is `rds-ca-rsa2048-g1`. Certificate authority date is `May 26, 2061, 05:04 (UTC+05:30)`. DB instance certificate expiration date is `September 18, 2026, 14:03 (UTC+05:30)`.

2. Secret Manger screen

The screenshot shows the AWS Secrets Manager console. A secret is listed with the following details:

Secret name	Description	Last retrieved (UTC)
<code>rdsdb-b583efca-6ece-46ae-81fd-06097b2e88a5</code>	The secret associated with the primary RDS DB instance: <code>arn:aws:rds:us-east-1:888342500922:db:database-1</code>	September 18, 2025

3. Connect the RDS database to MySQL using secret values from secrets manager

The screenshot shows the MySQL Workbench 'Setup New Connection' dialog. The connection name is 'dev', the method is 'Standard (TCP/IP)', and the host is `database-1.c2dg6mmy4rhh.us-east-1.rds.amazonaws.com`. The username is 'admin' and the password is stored in the vault. A success message is displayed:

Successfully made the MySQL connection

Information related to this connection:

- Host: `database-1.c2dg6mmy4rhh.us-east-1.rds.amazonaws.com`
- Port: `3306`
- User: `admin`
- SSL: `enabled with TLS_AES_128_GCM_SHA256`

A successful MySQL connection was made with the parameters defined for this connection.

Buttons: OK, Test Connection, Cancel, OK

4. Created database, created table in database, inserted records in database, retrieved records

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```
1 Create database devSnapshot;
2
3 Create Table devSnapshot.Employee(
4   id int,
5   name varchar(255),
6   address varchar(255),
7   city varchar(255)
8 );
9
10 Insert into devSnapshot.Employee(
11   id, name, address, city) values (
12   1, 'Yogesh', 'XYZ', 'Nagpur');
13 Insert into devSnapshot.Employee(
14   id, name, address, city) values (
15   2, 'Ramesh', 'PQR', 'Mumbai');
16
17 select * from devSnapshot.Employee;
```

The left sidebar shows the Schemas tree with 'devSnapshot' expanded, showing the 'Employee' table. The bottom pane shows the 'Result Grid' with the following data:

id	name	address	city
1	Yogesh	XYZ	Nagpur
2	Ramesh	PQR	Mumbai

The bottom pane also shows the 'Output' window with the following log:

#	Time	Action	Message
1	14:21:03	Create database devSnapshot	1 row(s) affected
2	14:22:38	Create Table Employee(id int, name varchar(255), address varchar(255), city varchar(255))	Error Code: 1046. No database selected Select the default DB to be used by double-clicking t
3	14:22:52	Create Table devSnapshot.Employee(id int, name varchar(255), address varchar(255), city varchar(255))	0 row(s) affected
4	14:24:36	Insert into devSnapshot.Employee(id, name, address, city) values (1, 'Yogesh', 'XYZ', 'Nagpur')	1 row(s) affected
5	14:24:44	Insert into devSnapshot.Employee(id, name, address, city) values (2, 'Ramesh', 'PQR', 'Mumbai')	1 row(s) affected
6	14:25:16	select * from devSnapshot.Employee LIMIT 0, 1000	2 row(s) returned

5. Created the Snapshot of database-1

The screenshot shows the 'Manual snapshots' page in the AWS Management Console. The page displays a table of manual snapshots with the following columns: Snapshot name, Engine version, DB instance or clus..., Snapshot creation time, and DB Instance created time. The table contains one snapshot:

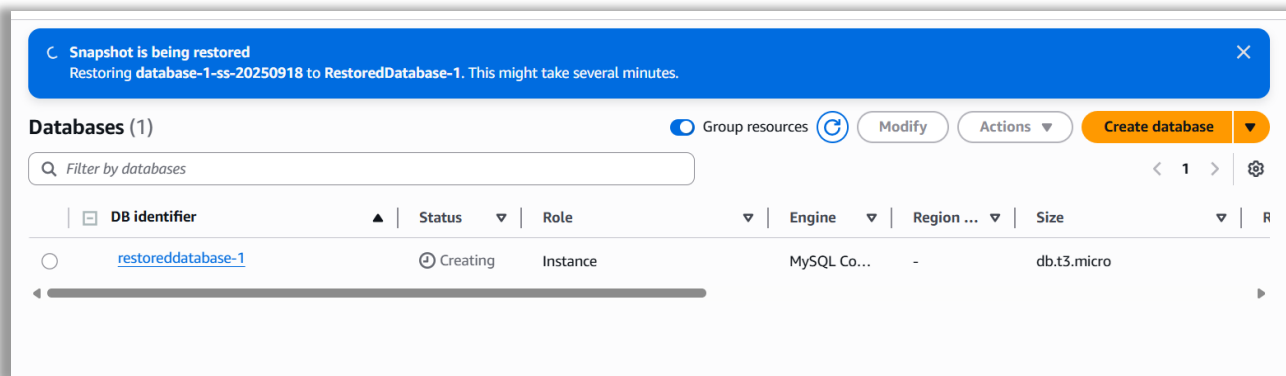
Snapshot name	Engine version	DB instance or clus...	Snapshot creation time	DB Instance created time
database-1-ss-20250918	8.0.42	database-1	September 18, 2025, 14:31 (UTC+05:30)	September 18, 2025, 14:05 (UTC+05:30)

6. Deleted the database-1 after taking snapshot

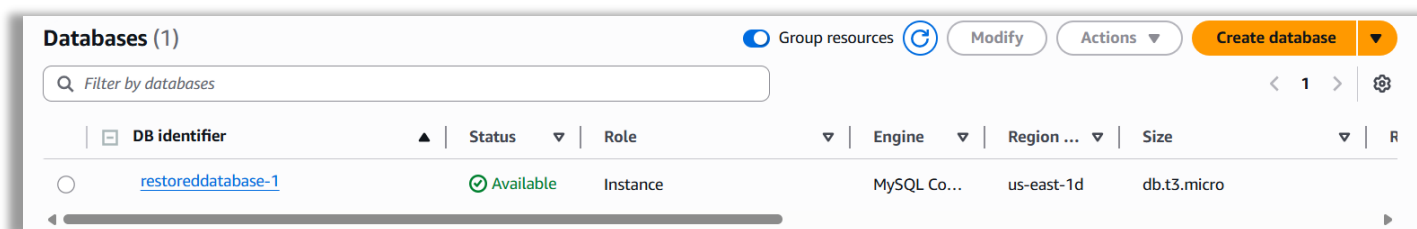
The screenshot shows the 'Deleting DB instance database-1' notification in the AWS Management Console. The notification bar is blue and contains the text 'Deleting DB instance database-1'. Below the notification bar, the 'Databases (1)' page is visible, showing a table of databases with the following columns: DB identifier, Status, Role, Engine, Region, and Size. The table contains one database:

DB identifier	Status	Role	Engine	Region	Size
database-1	Deleting	Instance	MySQL Co...	us-east-1a	db.t4g.micro

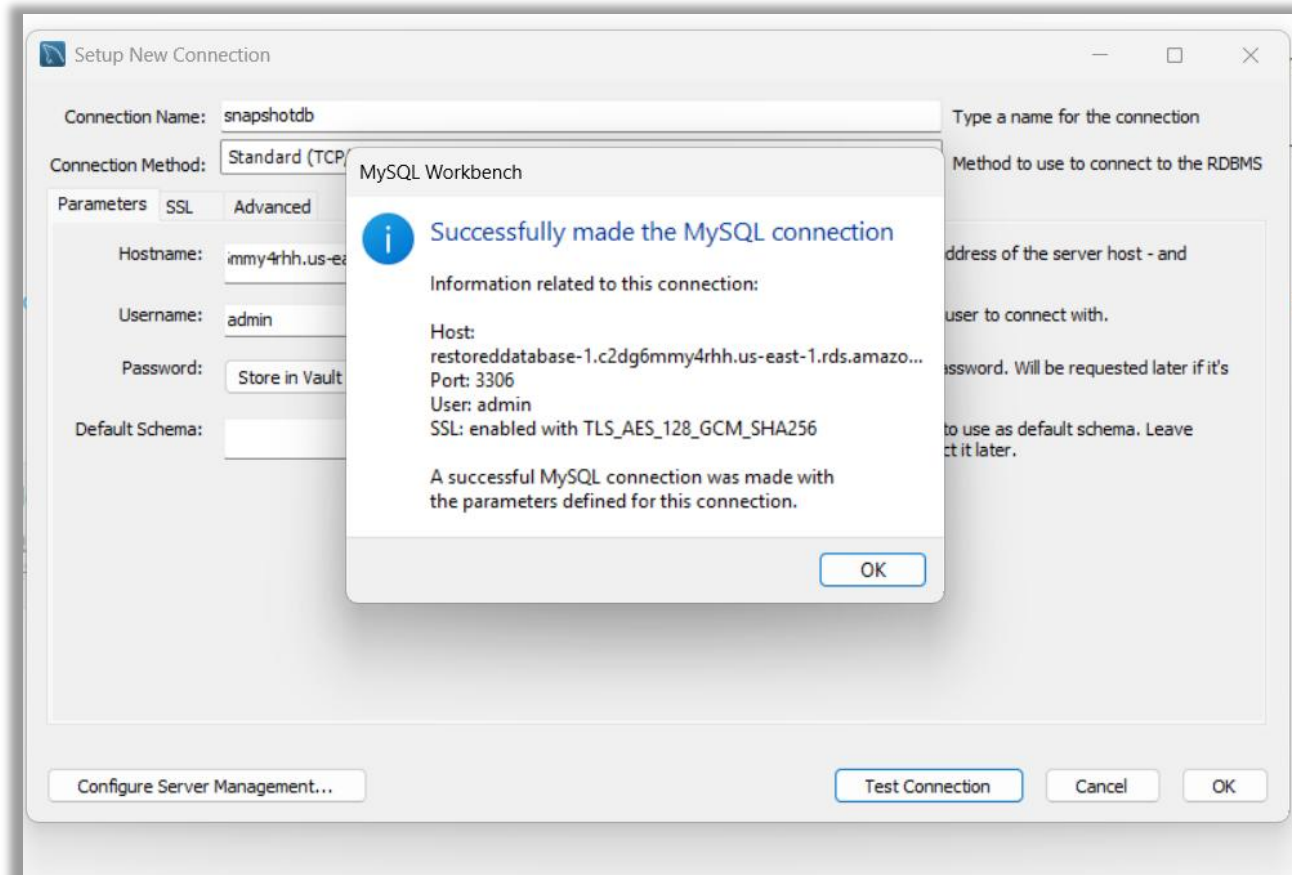
7. After deleted the existing database. Restoring it from the snapshot.



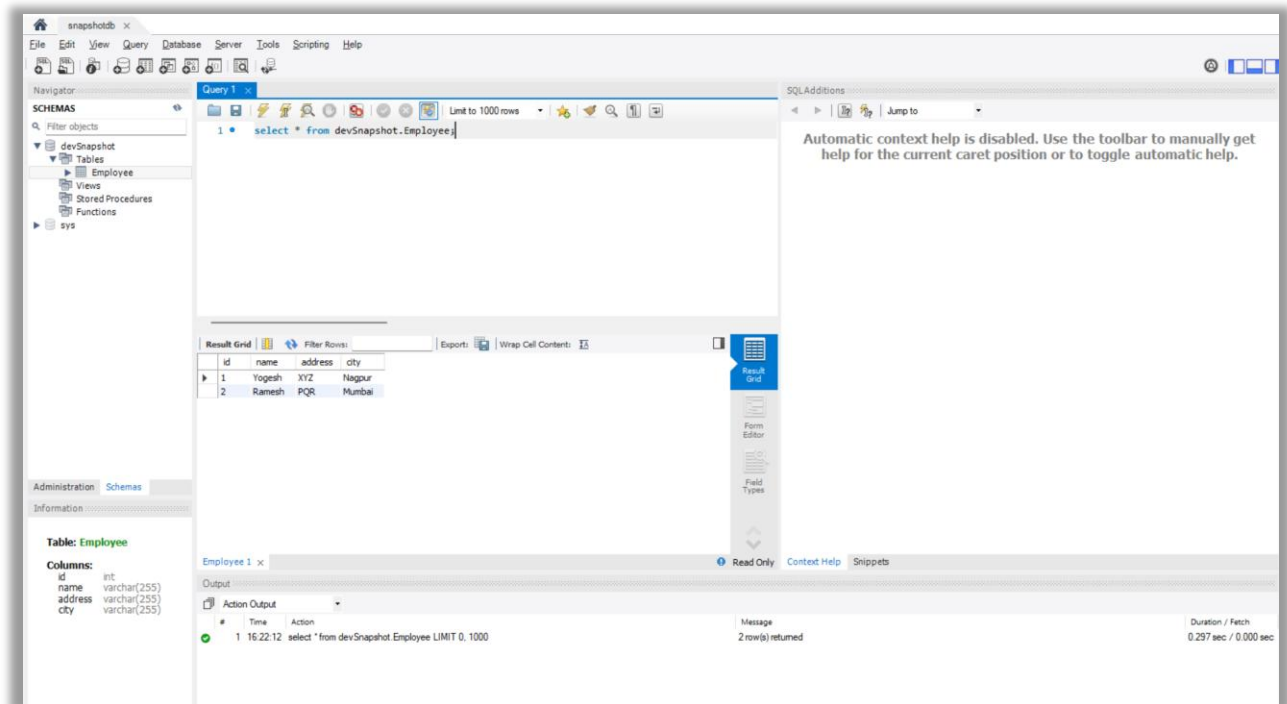
8. Snapshot is restored and database is available



9. The secret key was not restored automatically so modified the database to get the secret key again and then connect the MySQL;



10. Checked the connected database restored from snapshot and found the records that exists before database delete



Task 2:

Created a database with Private access (Publicly accessible : No)

database-1

Modify Actions

Summary

DB identifier database-1	Status Available	Role Instance	Engine MySQL Community	Recommendations
CPU 4.07%	Class db.t4g.micro	Current activity 0 Connections	Region & AZ us-east-1f	

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data

Connectivity & security

Endpoint & port Endpoint database-1.c2dg6mmy4rhh.us-east-1.rds.amazonaws.com Port 3306	Networking Availability Zone us-east-1f VPC vpc-06008718ac30863d8 Subnet group default-vpc-06008718ac30863d8 Subnets subnet-074100db9c88d1d1b subnet-0d52e0a0fa761063c subnet-0788c08d2a5c09f64	Security VPC security groups default (sg-070dc7952b1b5af5f) Active Publicly accessible No Certificate authority rds-ca-rsa2048-g1 Certificate authority date May 26, 2061, 05:04 (UTC+05:30) DB instance certificate expiration date
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2. Created EC2 Instance

Instance summary for i-078f43b5ea2c35dc1 (EC2ConnectMySQLTest) [Info](#)

Updated less than a minute ago

Instance ID

[i-078f43b5ea2c35dc1](#)

IPv6 address

–

Hostname type

IP name: ip-172-31-31-40.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

[174.129.49.68](#) [Public IP]

IAM Role

–

IMDSv2

Required

Operator

–

Public IPv4 address

[174.129.49.68](#) | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

[ip-172-31-31-40.ec2.internal](#)

Instance type

[t3.micro](#)

VPC ID

[vpc-06008718ac30863d8](#)

Subnet ID

[subnet-0d52e0a0fa761063c](#)

Instance ARN

[arn:aws:ec2:us-east-1:888342500922:instance/i-078f43b5ea2c35dc1](#)

Private IPv4 addresses

[172.31.31.40](#)

Public DNS

[ec2-174-129-49-68.compute-1.amazonaws.com](#) | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#)

Auto Scaling Group name

–

Managed

false

3. Connect the EC2 instance to run the Linux commands and Installed MariaDB for installing MySQL client

```
ec2-user@ip-172-31-31-40 ~]$ sudo su -
Last login: Fri Sep 19 13:47:06 UTC 2025 on pts/1
[root@ip-172-31-31-40 ~]# yum install mariadb105-server
Amazon Linux 2023 Kernel Livepatch repository                               204 kB/s | 23 kB    00:00
No match for argument: mariadb105-server
Error: Unable to find a match: mariadb105-server
[root@ip-172-31-31-40 ~]# yum install mariadb105-server
Last metadata expiration check: 0:01:17 ago on Fri Sep 19 13:59:48 2025.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
mariadb105-server	x86_64	3:10.5.29-1.amzn2023.0.1	amazonlinux	10 M
Installing dependencies:				
mariadb-connector-c	x86_64	3.3.10-1.amzn2023.0.1	amazonlinux	211 k
mariadb-connector-c-config	noarch	3.3.10-1.amzn2023.0.1	amazonlinux	9.9 k
mariadb105	x86_64	3:10.5.29-1.amzn2023.0.1	amazonlinux	1.5 M
mariadb105-common	x86_64	3:10.5.29-1.amzn2023.0.1	amazonlinux	28 k
mariadb105-errmsg	x86_64	3:10.5.29-1.amzn2023.0.1	amazonlinux	212 k
mysql-selinux	noarch	1.0.4-2.amzn2023.0.3	amazonlinux	36 k
perl-B	x86_64	1.80-477.amzn2023.0.7	amazonlinux	177 k
perl-DBD-MariaDB	x86_64	1.22-1.amzn2023.0.4	amazonlinux	153 k
perl-DBI	x86_64	1.643-7.amzn2023.0.3	amazonlinux	700 k
perl-Data-Dumper	x86_64	2.174-460.amzn2023.0.2	amazonlinux	55 k
perl-File-Copy	noarch	2.34-477.amzn2023.0.7	amazonlinux	20 k
perl-FileHandle	noarch	2.03-477.amzn2023.0.7	amazonlinux	15 k
perl-Math-BigInt	noarch	1:1.9998.39-2.amzn2023.0.2	amazonlinux	202 k
perl-Math-BigRat	noarch	0.2624-500.amzn2023.0.2	amazonlinux	42 k
perl-Math-Complex	noarch	1.59-477.amzn2023.0.7	amazonlinux	46 k
perl-Sys-Hostname	x86_64	1.23-477.amzn2023.0.7	amazonlinux	16 k
perl-base	noarch	2.27-477.amzn2023.0.7	amazonlinux	16 k
Installing weak dependencies:				
mariadb105-backup	x86_64	3:10.5.29-1.amzn2023.0.1	amazonlinux	6.0 M

4. Connected to RDS through MySQL client

```
Complete!
[root@ip-172-31-31-40 ~]# mysql -h database-1.c2dg6mmy4rhh.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 8.0.42 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases
-> ;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.010 sec)

MySQL [(none)]>
```

5. Database is created from command

```
MySQL [(none)]> create database testConnectRDS
-> ;
Query OK, 1 row affected (0.016 sec)

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql      |
| performance_schema |
| sys       |
| testConnectRDS  |
+-----+
5 rows in set (0.004 sec)

MySQL [(none)]>
```

6. Given all data to try to connect through Workbench :

Setup New Connection

Connection Name: Type a name for the connection

Connection Method: Method to use to connect to the RDBMS

Parameters SSL Advanced

SSH Hostname: SSH server hostname, with optional port number.

SSH Username: Name of the SSH user to connect with.

SSH Password: Clear SSH user password to connect to the SSH tunnel.

SSH Key File: ... Path to SSH private key file.

MySQL Hostname: MySQL server host relative to the SSH server.

MySQL Server Port: TCP/IP port of the MySQL server.

Username: Name of the user to connect with.

Password: Clear The MySQL user's password. Will be requested later if not set.

Default Schema: The schema to use as default schema. Leave blank to select it later.

Configure Server Management... Test Connection Cancel OK

7. Connected through Workbench:



Setup New Connection

Connection Name: TestPrivateRDSConn

Type a name for the connection

Connection Method: Standard TCP/IP

Method to use to connect to the RDBMS

MySQL Workbench

Parameters SSL Advanced

SSH Hostname: 54.197.198

hostname, with optional port number.

SSH Username: ec2-user

SSH user to connect with.

SSH Password: Store in

password to connect to the SSH tunnel.

SSH Key File: C:\Users\yj

private key file.

MySQL Hostname: database-1

or host relative to the SSH server.

MySQL Server Port: 3306

of the MySQL server.

Username: admin

user to connect with.

Password: Store in Vault ...

Clear

The MySQL user's password. Will be requested later if not set.

Default Schema:

The schema to use as default schema. Leave blank to select it later.



Successfully made the MySQL connection

Information related to this connection:

Host:

database-1.c2dg6mmy4rhh.us-east-1.rds.amazonaws.com

Port: 3306

User: admin

SSL: enabled with TLS_AES_128_GCM_SHA256

A successful MySQL connection was made with the parameters defined for this connection.

OK

Configure Server Management...

Test Connection

Cancel

OK